SIMATS ENGINEERING

CSA0705 - COMPUTER NETWORKS

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NATURE OF ASSIGNMENT : INDIVIDUAL

1. Given Data

The smart manufacturing plant has four traffic sources. The calculations below use decimal networking units: 1 Mbps = 1,000,000 bits per second and 1 Gbps = 1,000 Mbps.

Source	Quantity	Traffic details
Robot controllers	40	200-byte command every 2 ms per controller
Vibration sensors	600	50 bytes per second per sensor
CCTV cameras	24	4 Mbps per camera, continuous
Office users	150	2 Mbps per active user; 10% active at peak

2. Traffic Calculation for One Production Line

A. Robot controllers

Each controller sends 1 command every 2 ms. Therefore, commands per second per controller = 1000 ms / 2 ms = 500 commands/s.

Traffic per controller = 200 bytes × 500 × 8 bits/byte = 800,000 bits/s = 0.8 Mbps.

For 40 controllers = 40 × 0.8 = **32 Mbps**.

B. Vibration sensors

Traffic = 600 sensors × 50 bytes/s × 8 bits/byte
= 240,000 bits/s
= **0.24 Mbps**.

C. CCTV cameras

Traffic = 24 cameras × 4 Mbps per camera
= **96 Mbps**.

D. Office users

Active users = 150 × 10/100 = 15 users.

Traffic = 15 × 2 Mbps = **30 Mbps**.

Traffic source	Calculation	Traffic
Robot controllers	40 × 0.8 Mbps	32 Mbps
Vibration sensors	600 × 50 × 8 bits/s	0.24 Mbps
CCTV cameras	24 × 4 Mbps	96 Mbps
Office users	15 × 2 Mbps	30 Mbps
Total	**32 + 0.24 + 96 + 30**	**158.24 Mbps**

3. Traffic for Eight Production Lines

Assuming all eight lines generate the same traffic:

Total plant traffic = 158.24 Mbps × 8

= **1,265.92 Mbps** (approximately **1.266 Gbps**).

4. Highest Data Source and Strictest Delay Requirement

The CCTV cameras generate the most traffic at **96 Mbps** per line. The robot controllers have the strictest delay requirement because robot commands must arrive within **10 ms**. Therefore, the source generating the most data and the source with the strictest delay requirement are **not the same**.

5. Is a Single 1 Gbps Uplink Sufficient?

For one line: 158.24 Mbps is less than 1,000 Mbps, so the link has enough nominal capacity for the calculated traffic of one line.

For eight lines: 1,265.92 Mbps is greater than 1,000 Mbps, so a single 1 Gbps uplink is **not sufficient** for the calculated traffic of all eight lines.

Effect of a shared queue: If CCTV packets and robot-control packets use the same first-in, first-out queue, a burst of video packets can make a robot command wait behind them. This queueing delay could threaten the robot's 10 ms deadline. The network should use Quality of Service (QoS) so that time-critical robot-control traffic receives higher priority than ordinary video and office traffic. Capacity planning and queue management are both necessary.

6. Ranking: Effect of a 100 ms Delay

The assignment does not give exact delay limits for sensors, CCTV or office traffic. The following ranking is therefore a reasoned engineering judgement based on the plant scenario.

Rank	Traffic type	Reason
1	Robot commands	A 100 ms delay is ten times the stated 10 ms command deadline and may disrupt control.
2	Vibration sensors	Delayed readings can postpone detection of abnormal vibration or developing machine faults.
3	CCTV video	A 100 ms delay affects live viewing and operator awareness, but recorded video can still be useful.
4	Office / ERP / e-mail	A short delay is generally less harmful to routine office work than to industrial control.

7. Alternative Concept: Local CCTV Recording

If cameras record locally and transmit motion-triggered clips for about 5% of the time, the average CCTV network traffic is reduced to 5% of its continuous-streaming value.

New CCTV traffic per line = 96 Mbps × 5/100

= **4.8 Mbps**.

New total traffic per line = 32 + 0.24 + 4.8 + 30

= **67.04 Mbps**.

New total for eight lines = 67.04 × 8

= **536.32 Mbps**.

536.32 Mbps < 1,000 Mbps: the 1 Gbps uplink is sufficient for the calculated average traffic.

This result assumes that 5% represents the average fraction of the original CCTV bit rate transmitted over the network, and that all eight lines have the same traffic profile. Actual peak traffic, protocol overhead and spare-capacity requirements should also be considered.

8. Final Conclusion

The calculated traffic of one production line is **158.24 Mbps**, and the traffic for eight lines is **1,265.92 Mbps**. Therefore, one 1 Gbps uplink can carry the calculated traffic of one line but cannot carry the calculated traffic of all eight lines. CCTV produces the highest traffic (96 Mbps per line), while robot commands have the strictest delay requirement (10 ms). A shared queue could delay robot commands, so QoS priority should be used for control traffic. If CCTV records locally and sends motion-triggered clips for 5% of the time, the calculated eight-line traffic falls to **536.32 Mbps**, making a 1 Gbps uplink sufficient for the average calculated load, subject to peak-load and overhead checks.